

A Telepresence System using Toy Robots the Users can Assemble and Manipulate with Finger Plays and Hand Shadow

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ABSTRACT

The authors report a telepresence system using toy robots. The robots can be assembled and set up for telepresence by the user easily. The robots are manipulated with hand gestures of finger play and hand shadow without instruction, since the users already know these gestures by their experience. Through the experiment, the system was found to be easy to use, and the users felt familiar with the robot and didn't feel nervous.

Keywords: Telepresence, gesture interface, hand shadow, finger play.

Index Terms: Human-centered computing — Human computer interaction (HCI) — Interaction paradigms — Collaborative interaction

1 INTRODUCTION

Communication using avatars such as VRChat and [1] has been developed. In general, there are problems to make this kind of communication more enjoyable: difficulty in customizing the avatar, limitation in the form of the avatar, and burdens and costs in motion capture. In this paper, the authors developed a telepresence system with toy robots using the method reported in [2, 3]. With this method, the user manipulates toy robots by hand gestures of finger plays and hand shadow. Through the experiments with the implemented system, the authors show: 1) the users can easily assemble and set up their own robots for telepresence; 2) the burden of operation is reduced; 3) the other party feels familiar to the robot and doesn't feel nervous.

2 HAND GESTURE OF FINGER PLAYS AND HAND SHADOW

The method adopted in this paper is to manipulate 3D models and (toy) robots by hand gesture of finger plays and hand shadow. In this method, the user makes the shape of the 3D model or the robot to manipulate with the hands. For example, when manipulating a crocodile, the user makes its shape with the hands. The crocodile (3D model or robot) opens and closes its mouth by the user's extending and flexing the right thumb. It wags the tail with the motion of his/her left thumb. The users can find out how to manipulate the 3D model or the robot by themselves with their knowledge and experience of finger plays and hand shadow, without instruction. Since various animals and objects can be represented with finger plays and hand shadow, various models and robots are manipulated by the proposed method. The direct feeling of manipulation is also one of its features.

Prior to use, the user's fingers are to be associated with joints of the 3D model or the robot. For example, the angle between the right



Figure 1: The system at work; the operator manipulates the robot with hand gestures and communicates with the partner

thumb and index finger is associated with the angle of the crocodile's mouth (with range adjustment, if needed). The directions of the fingers are also taken into account to realize various movements. Since these associations are specific to each model or robot, configuration is required before use.

3 SYSTEM IMPLEMENTATION

The system works on Unity (for Windows and macOS) with a Leap Motion sensor. PCs are to be connected to the network. ArtecRobo Advance (toy blocks and motors) is used to build toy robots. The system has the following functions: 1) The robot can be remote-controlled by hand gestures; 2) The video and audio from the robot is displayed to the operator; 3) The operator's voice is heard via the robot; 4) The robot can be assembled by the user.

The operator manipulates the robot by moving the hands above the Leap Motion sensor. The crocodile robot (in Figure 1) opens and closes its mouth by the user's right hand as described in Section 2, and wags its tail by the user's bending and stretching the left thumb. These motions are based on a hand shadow of a crocodile. In addition, the robot itself can move when the wheels and motors are attached.

The robot equips a webcam, a microphone, and a speaker, so that the operator can communicate with the other party by video and audio.

When the robot is assembled by the user him/herself, the procedure of settings is as follows: 1) Assembling the toy robot with blocks, 2) connecting the joints to move with the motors, 3) attaching the wheels and motors, 4) associating the motions of the robots with motions of the user's hands, 5) configuring how to move the robot itself, and 6) attaching a webcam, a microphone, and a speaker to the robot.

The motions of the robot (step 4) and 5), described above) can be set up with the GUI (Figure 2). In this process, the system recognizes the user's motion demonstrated above the Leap Motion sensor, and associates the robot's motion with the user's demonstrated motion.

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Figure 2: The GUI for configuration of the robot's motion

4 EXPERIMENT OF TELEPRESENCE COMMUNICATION

4.1 Overview of the Experiment

Six right-handed males aged 22–26 years old participated in the experiment, in pairs: one of them operated the crocodile robot and the other was the partner (human), and they communicated in a separate room. After the experiment, the participants answered a questionnaire.

4.2 Results of the Experiment

The operators of the crocodile robot talked with the partner after the instruction. All participants were able to control the mouth and tail of the robot, and move the robot itself. Some participants played with the robot when they communicated as the partner.

As for the questionnaire about the crocodile robot, most of the participants said that they didn't feel nervous and that they felt expressiveness of the robot and presence through the robot. As for the questionnaire about the other party (human), all participants felt familiar with the robot, most of them answered that there was no problem in the communication. And, the delay of communication wasn't a problem for them.

5 EXPERIMENT OF ASSEMBLING TELEPRESENCE ROBOTS

The experiment to assemble a robot was also carried out for three of the participants in the experiment in section 4. The participants assembled and configured the robots according to the procedure described in section 3. They were able to configure the robots by themselves after the instruction. After configuration, they checked the setting the robots and operate the robots from another room.

In the experiment, the participants operated the robots as asked by the authors. They were able to operate the robot well enough, and it was confirmed that the users could immediately use the assembled robots for telepresence.

The assembled robots were a monkey, a fowl, and a peacock (Figure 3). For example, the user could bend and extend the monkey's left/right foreleg with the user's left/right hand, respectively. The user could also move the monkey forward and backward by the user's bending and stretching the arms.

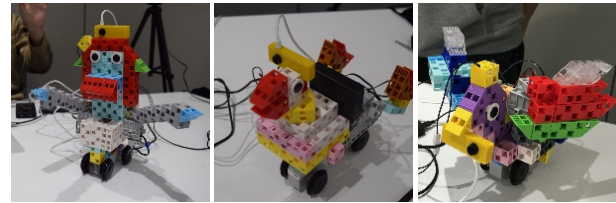


Figure 3: The toy robots the participants assembled (a monkey, a fowl, and a peacock)

6 DISCUSSION

In the telepresence communication experiment, the operator was able to remote-control the robot appropriately to communicate with his partner. The evaluations of the system were mostly positive by both the operator and the partner.

There were the scenes that the partner played with the robot, which reflected the fact that the participants said that they felt familiar with the robot and didn't feel nervous.

It was also confirmed that the telepresence robot could be easily assembled, set up, and operated. It can be said that the telepresence system has high usability and that it is operated at low cost.

7 CONCLUSION AND FUTURE WORKS

The authors have implemented a telepresence system using toy robots. The method to manipulate robots is based on the finger plays and hand shadow. In the evaluation experiment, the participants said that they felt familiar with the robot and that they were able to interact with the robot light-heartedly. In addition, the users assembled and set up the robots for telepresence easily.

Improving the quality of the video and audio is one of the future works. This is because the voice was sometimes difficult to hear in the experiment. And improving the sense of presence by using an omnidirectional camera and HMD is interesting.

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